# script.js

```
// ---------- Data ----------
const skills = [
 { name: "HTML5", level: 90 }, { name: "CSS3 & Responsive Design", level: 85 },
 { name: "JavaScript (ES6)", level: 75 }, { name: "Python", level: 80 },
 { name: "Git & GitHub", level: 70 }, { name: "Machine Learning Basics", level: 65 }
];
const projects = [
 { title: "TastyBites Food Ordering", cat: "Web", tech: ["HTML", "CSS", "JavaScript"],
 desc: "Online food ordering website with menu search, cart and order tracking.",
 code: "https://github.com/<your-username>/online-food-ordering", demo: "#" },
 { title: "Interactive Quiz App", cat: "Web", tech: ["HTML", "JavaScript"],
 desc: "Quiz application with timer, score calculation and result summary.",
 code: "https://github.com/<your-username>/quiz-app", demo: "#" },
 { title: "Student Marks Predictor", cat: "AI / ML", tech: ["Python", "Scikit-learn"],
 desc: "Predicts a student's final marks from study hours using linear regression.",
 code: "https://github.com/<your-username>/marks-predictor", demo: "#" },
 { title: "Sentiment Analyzer", cat: "AI / ML", tech: ["Python", "NLTK"],
 desc: "Classifies product reviews as positive, negative or neutral.",
 code: "https://github.com/<your-username>/sentiment-analyzer", demo: "#" },
 { title: "Expense Tracker", cat: "Python", tech: ["Python", "SQLite"],
 desc: "Command-line tool to record, search and summarise daily expenses.",
 code: "https://github.com/<your-username>/expense-tracker", demo: "#" },
 { title: "Student Portfolio Website", cat: "Web", tech: ["HTML", "CSS", "JavaScript"],
 desc: "This responsive portfolio that presents my profile, skills and projects.",
 code: "https://github.com/<your-username>/student-portfolio", demo: "#" }
];
let activeCat = "All";
const $ = (id) => document.getElementById(id);

// ---------- Navigation ----------
$("menuBtn").addEventListener("click", () => $("nav").classList.toggle("open"));
document.querySelectorAll(".nav-link").forEach(a =>
 a.addEventListener("click", () => $("nav").classList.remove("open")));

// ---------- Typing animation ----------
const roles = ["Web Developer", "AI Enthusiast", "Problem Solver", "Quick Learner"];
let r = 0, c = 0, deleting = false;
function typeEffect() {
 const word = roles[r];
 $("typed").textContent = word.slice(0, deleting ? --c : ++c);
 if (!deleting && c === word.length) { deleting = true; setTimeout(typeEffect, 1200);
     return; }
 if (deleting && c === 0) { deleting = false; r = (r + 1) % roles.length; }
 setTimeout(typeEffect, deleting ? 50 : 100);
}

// ---------- Skills Module ----------
function renderSkills() {
 $("skillList").innerHTML = skills.map(s => `
 <div class="skill"><div
     class="skill-top"><span>${s.name}</span><span>${s.level}%</span></div>
 <div class="bar"><div class="fill" data-level="${s.level}"></div></div></div>`).join("");
}

// ---------- Projects Module ----------
function renderFilters() {
 const cats = ["All", ...new Set(projects.map(p => p.cat))];
 $("filters").innerHTML = cats.map(c =>
 `<button class="chip ${c === activeCat ? "active" : ""}"
     data-cat="${c}">${c}</button>`).join("");
}
function renderProjects() {
 const list = projects.filter(p => activeCat === "All" || p.cat === activeCat);
 $("projectGrid").innerHTML = list.map(p => `
 <div class="card">
 <span class="cat">${p.cat}</span><h3>${p.title}</h3><p>${p.desc}</p>
 <div>${p.tech.map(t => `<span class="tag">${t}</span>`).join("")}</div>
 <div class="links"><a href="${p.code}" target="_blank">Source Code</a>
 <a href="${p.demo}" target="_blank">Live Demo</a></div>
 </div>`).join("");
}
// ---------- Contact Module (validation) ----------
```

```
const emailPattern = /^[^\s@]+@[^\s@]+\.[a-zA-Z]{2,}$/;
function setError(id, msg) {
 $(id + "Err").textContent = msg;
 $(id).classList.toggle("invalid", msg !== "");
 return msg === "";
}
function validateForm(e) {
 e.preventDefault();
 const name = $("name").value.trim(), email = $("email").value.trim(), msg =
     $("message").value.trim();
 $("formSuccess").textContent = "";
 let ok = setError("name", name ? "" : "Name is required.");
 ok = setError("email", !email ? "Email is required." :
 emailPattern.test(email) ? "" : "Enter a valid email address.") && ok;
 ok = setError("message", msg.length >= 10 ? "" : "Message must have at least 10
     characters.") && ok;
 if (ok) {
 $("formSuccess").textContent = `Thank you, ${name}! Your message has been sent.`;
 $("contactForm").reset();
 }
}

// ---------- Scroll effects ----------
const observer = new IntersectionObserver(entries => entries.forEach(en => {
 if (!en.isIntersecting) return;
 en.target.classList.add("show");
 en.target.querySelectorAll(".fill").forEach(f => f.style.width = f.dataset.level + "%");
}), { threshold: 0.15 });

window.addEventListener("scroll", () => {
 $("toTop").classList.toggle("show", window.scrollY > 400);
 document.querySelectorAll("main section, #home").forEach(sec => {
 if (window.scrollY + 120 >= sec.offsetTop)
 document.querySelectorAll(".nav-link").forEach(a =>
 a.classList.toggle("active", a.getAttribute("href") === "#" + sec.id));
 });
});

// ---------- Events ----------
document.addEventListener("click", (e) => {
 if (e.target.dataset.cat) { activeCat = e.target.dataset.cat; renderFilters();
     renderProjects(); }
});
$("toTop").addEventListener("click", () => window.scrollTo({ top: 0, behavior: "smooth" }));
$("contactForm").addEventListener("submit", validateForm);

renderSkills(); renderFilters(); renderProjects(); typeEffect();
document.querySelectorAll(".reveal").forEach(s => observer.observe(s));
```